Trustworthiness is not one property: accuracy, cross-dataset robustness, explainability, abstention, and deployment cost in single-cell cervical cytology classification

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Abstract. Purpose: Lightweight vision transformers are proposed for automated cervical cytology screening, but "trustworthy AI" is usually evaluated as one property that a pipeline has or lacks. We tested whether accuracy, cross-dataset robustness, explainability, selective prediction, and deployment cost agree with one another on one single-cell cervical cytology classification task, using a common model registry but different backbones for different properties. **Approach:** We benchmarked 12 convolutional, full-size transformer, and efficient transformer models on SIPaKMeD (five classes) and evaluated all 12 zero-shot on three external cytology datasets by mapping their outputs to a binary normal/abnormal label: Herlev (single-cell crops, the primary external test) and Mendeley LBC and APCEM (multi-cell fields of view, the latter algorithmically enhanced, treated as content-shift stress tests), so the external evaluation combines differences in institution, scale, content, preprocessing, and annotation. On a standard vision transformer, we compared three explainability methods against deletion/insertion faithfulness and expert-annotated nucleus boundaries, tested a combined confidence and conformal-set-size abstention rule, and measured parameters, floating-point operations (FLOPs), central processing unit (CPU) latency, and explanation cost. **Results:** The two smallest models (about 2.1M parameters) were the two most accurate in distribution in the primary run and ranked second and third by three-seed mean, on a cell-level split in which most test cells share a source image with training cells, which could inflate in-distribution scores; a cluster-grouped re-run changed balanced accuracy by a median of -0.9 points across the 12 models (range -14.6 to $+1.1$, the extreme being a training failure), comparable to seed variation. In-distribution accuracy did not predict robustness consistently: the in-distribution leader ranked 10th of 12 on both Herlev and Mendeley LBC in the primary run, and on APCEM (multi-cell fields) 9 of 12 models fell below chance on the area under the receiver operating characteristic curve. Rank correlations between three-seed mean external accuracy and in-distribution accuracy, parameter count, and FLOPs were weak to moderate and changed sign across datasets, and model rankings changed across training seeds and datasets. Relevancy propagation was the most faithful explainability method on all three metrics on two backbones, with paired cluster-bootstrap intervals excluding zero in all 12 comparisons over 596 test cells (nucleus concentration 3.8 times chance versus 1.3 for attention rollout on DeiT-Tiny), although concentration on the nucleus does not establish clinical validity. Combining confidence with conformal set size helped only under severe shift (risk-coverage area under the curve 0.286 versus 0.324) and hurt in distribution and under mild shift. FLOPs did not predict CPU latency, and an explanation cost about 3.3 times a bare prediction. **Conclusions:** Accuracy, robustness, explainability, abstention, and efficiency resolved differently across experiments on the same task, so each needs its own test under the conditions of intended deployment; whether a joint test on a single backbone would change these conclusions was not tested.

Keywords: cervical cytology, vision transformers, cross-dataset generalization, explainability, selective prediction, model efficiency.

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1 Introduction

Cervical cancer remains a significant cause of cancer morbidity and mortality worldwide[1] and is substantially more curable when detected at a localized stage, yet access to screening and specialist cytology capacity are limited in many settings. Vision transformers (ViTs), often marketed as efficient or lightweight variants, are proposed as classification backbones for automated cytology screening. Such a tool is only useful if it fails visibly rather than silently.

"Trustworthy AI" is usually evaluated as a single property, but a screening classifier makes at least five separable claims: a lightweight model gives up little accuracy; its accuracy transfers to other datasets; its explanations are faithful; its confidence supports a reliable abstention rule; and its efficiency claim holds under real measurement. We test them side by side on one single-cell cervical cytology task and ask whether they resolve alike. The experiments share a dataset family and a 12-model registry but not a single backbone (Sec. 6): the benchmark covers 12 models, the explainability and abstention experiments use DeiT-Tiny, and the degradation experiment uses EfficientViT-M0.

Our contributions are (i) a 12-model benchmark on SIPaKMeD spanning convolutional neural network (CNN), full-size ViT, and efficient ViT families; (ii) a zero-shot evaluation of all 12 models on three external datasets, with a three-seed sensitivity check and a cluster-grouped re-run, showing that in-distribution accuracy, parameter count, floating-point operations (FLOPs), and architecture family did not predict robustness consistently across datasets; (iii) a paired comparison of relevancy propagation, attention rollout, and Grad-CAM against behavioral faithfulness and expert nucleus boundaries, and a test of whether combining temperature-scaled confidence with conformal set size improves abstention; and (iv) a measurement of parameters, FLOPs, central processing unit (CPU) latency, and explanation cost. The properties resolve differently across experiments, which shows that they are not interchangeable but not that a joint test on a single backbone is necessary to see this.

2 Related Work

2.1 Efficient Vision Transformers

Efficient transformers include hybrid local–global designs (MobileViT[2], MobileViTv2[3]), distillation-oriented designs (DeiT[4], built on the ViT backbone[5]), and hardware-aware designs (LeViT[6], TinyViT[7]). EfficientViT-M0 (cascaded group attention[8]) and EfficientViT-B0 (multi-scale linear attention[9]) are unrelated designs despite sharing a name and a `timm` tag prefix. These papers do not evaluate cross-dataset shift as part of their efficiency claims, and parameter count, FLOPs, and measured speed are known to disagree as cost indicators,[2,10,11] although we found no study of this for cytology models.

2.2 *Cross-Dataset Shift and Data Splitting*

Models selected on held-out in-distribution data can degrade under real shift across scanners, sites, and populations,[12,13] and in-distribution accuracy can be a poor guide because models that perform equivalently on held-out data behave differently once the data change.[14–16] Predictive uncertainty also degrades under shift.[17] In cervical cytology, recent studies have examined classifier behavior on real-world and multi-scanner data,[18,19] and external validation across independent datasets has been described as rare.[20] Splitting by image, tile, or record when several samples come from one subject can inflate accuracy relative to subject-level splitting in digital pathology,[21] clinical machine learning,[22] and brain MRI.[23] For SIPaKMeD, Albzour and Lam note that cropped-cell filenames encode the parent cluster image and that parent-level grouping was not enforced in their cross-validation, a limitation they did not measure;[20] we measure the overlap under our split (Sec. 3.2.1) and gauge its effect with a cluster-grouped re-run (Sec. 4.3).

2.3 *Explainability, Selective Prediction, and Adjacent Work*

Attention rollout[24] treats chained attention weights as relevance, a premise contested in the literature on whether attention is an explanation;[25,26] relevancy propagation[27] weights attention by class gradients, and Grad-CAM[28] is commonly adapted to transformers through token activations. Deletion/insertion faithfulness[29] measures the effect of removing or revealing the regions an explanation marks as important. Prior studies have evaluated explanation methods for medical ViTs, including Grad-CAM and attention-based methods;[30–32] in our search we did not find work that scores explanations against expert nucleus masks in cervical cytology. Temperature scaling[33] and split conformal prediction[34] give calibrated confidence and a coverage-guaranteed set whose size is a coarse uncertainty proxy. Conformal prediction has been applied to cervical atypia classification[38] and calibration with selective prediction to Pap smear classification,[39] and we found no work combining confidence with conformal set size for abstention under cross-dataset shift in cytology. MC Dropout,[35] deep ensembles,[36] and evidential learning[37] are alternative signals we do not evaluate. Comparisons of many CNN and transformer architectures on cytology images,[40] EfficientViT ensembles on SIPaKMeD and Herlev,[41] and systematic ViT evaluations for cervical classification[42] exist; judging from titles and abstracts, none evaluates zero-shot transfer across independent datasets. Adjacent work includes cervical cell classification with hybrid feature fusion[43] and a stacking ensemble with explainable AI,[44] a lightweight transformer for plant disease[45] that motivates treating efficiency and explainability as joint goals, and explainable clinical AI in other domains.[46,47] These statements reflect a non-exhaustive search in October 2026 (Crossref, arXiv, web search); absence of a prior study in our search does not establish that none exists.

3 Materials and Methods

3.1 Datasets

All data are publicly available, previously published images, and no new patient data were collected. Models were trained on SIPaKMeD[48]: 4,049 native-resolution single-cell crops derived from the dataset's cytoplasm outlines, in five cytomorphology classes. External evaluation used Herlev,[49] Mendeley LBC,[50] and APCEM[51] (4,253 images in six Bethesda-based classes). Because the taxonomies differ, each model's five-class softmax output was summed into normal and abnormal probabilities under the label mapping in Table S9. The datasets also differ in what an image contains (Table 1): SIPaKMeD and Herlev images are single-cell crops, Mendeley LBC and APCEM images show many cells in a field of view, and the dataset authors enhanced APCEM images algorithmically. All images were resized without cropping to each model's square input, which changes the aspect ratio of non-square images. A single-cell classifier is therefore applied to whole fields in two of three external evaluations. We treat Herlev, which preserves the single-cell task, as the primary external test and Mendeley LBC and APCEM as content-shift stress tests, not controlled cross-institution tests. Because the datasets also differ in label definitions and their documentation does not establish category correspondences, "domain shift" here means combined differences in institution, scale, content, preprocessing, and annotation. The external sets are imbalanced (abnormal: Herlev 74%, APCEM 82%, Mendeley LBC 36%; SIPaKMeD 40%), so balanced accuracy is reported.

Table 1: What an image contains in each dataset. Image sizes were measured on a random sample of 60 images per dataset (SIPaKMeD sizes depend on the cytoplasm outline and were not tabulated). All images were resized to the model's square input.

Dataset	Image content	Image size (pixels)	Preprocessing noted
SIPaKMeD	Single-cell crop	Variable (cytoplasm outline plus 4-pixel margin)	Cropped by us
Herlev	Single-cell crop	Median 123×108 (range 43–331 by 53–363)	None noted
Mendeley LBC	Field of view with many cells (liquid-based cytology)	All sampled images 2048×1536	None noted
APCEM	Field of view with many cells	All sampled images 224×224	Enhanced by the dataset authors (APCEM procedure)

3.2 Models and Training

Twelve models spanning three families were evaluated: CNNs (ResNet-50,[52] MobileNetV2,[53] EfficientNet-Lite0[54]), full-size ViTs (ViT-Small,[5] Swin-Tiny[55]), and efficient ViTs (DeiT-Tiny,[4]

LeViT-128,[6] TinyViT-5M,[7] MobileViT-S,[2] MobileViTv2,[3] EfficientViT-B0,[9] EfficientViT-M0[8]). All `timm` tags were verified to exist with pretrained ImageNet weights, and the EfficientViT lineages were confirmed against `timm`'s model cards. Models were initialized from ImageNet weights and trained on five classes with AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), a cosine schedule over 30 epochs, cross-entropy loss, batch size 32, and early stopping on validation loss (patience 7), keeping the lowest-validation-loss checkpoint. Input size followed each model's `timm` configuration, with random flips, rotation up to 20 degrees, and brightness and contrast jitter of 0.15. We report balanced accuracy, macro F1, and the area under the receiver operating characteristic curve (AUROC). Five-class balanced accuracy has a chance level of 0.20 and binary external balanced accuracy 0.50, so they are compared by rank only.

3.2.1 Data split and source-image overlap

SIPaKMeD was split 60/15/10/15 into training, validation, calibration, and test sets, stratified by class and at the level of single cells (seed 13 plus the class index). The 4,049 cells come from 966 cluster images (about 4.2 per image), and cells of one cluster image were not kept together: reproducing the split on the crop filenames shows that 533 of 607 test cells (87.8%) share a cluster image with a training cell. Patient and slide identifiers are unavailable. In-distribution results, including calibration, abstention, and degradation results, may therefore be optimistic, and Sec. 4.3 gauges the effect with a cluster-grouped re-run. External evaluations are unaffected, because no external image was used for training, validation, or calibration. The same split manifest was used at all seeds.

3.3 Cross-Dataset Evaluation and Seeds

Each model was evaluated zero-shot on Herlev, Mendeley LBC, and APCEM, without retraining or recalibration. To measure dependence on the training run, all 12 models were retrained at two further seeds (17 and 29) and evaluated on all three external datasets from a single checkpoint per model per seed.

3.4 Explainability Methods and Metrics

Relevancy propagation and attention rollout are defined for standard multi-head self-attention blocks, and EfficientViT-M0 has no established propagation rule for either, so the explainability and abstention experiments use DeiT-Tiny (0.9606 balanced accuracy in the benchmark), with ViT-Small as a second plain-transformer backbone. All methods produce a relevance map over the 14×14 patch grid, upsampled to image resolution. *Attention rollout* chains head-averaged attention across the 12 blocks and is class-agnostic. *Relevancy propagation* uses the same recurrence with each block weighted by $\mathrm{ReLU}(\nabla_{\mathrm{attn}} \odot \mathrm{attn})$ for the predicted class and is class-discriminative. *Grad-CAM* is applied to the second-to-last block's token activations: DeiT-Tiny pools only the classification token, so patch-token gradients at the last block's output are exactly zero (verified

directly), and targeting that block would silently give an all-zero heatmap. Faithfulness is scored by deletion and insertion area under the curve (AUC),[29] replacing or revealing patches in order of relevance from a heavily blurred copy of the image (lower is better for deletion, higher for insertion). Localization is scored by the nucleus concentration ratio: the fraction of relevance mass inside SIPaKMeD's expert nucleus polygon divided by the nucleus's share of the image, where 1.0 is chance.

3.5 Calibration, Conformal Prediction, and Abstention

Temperature scaling was fit on a calibration split disjoint from training and test data. Conformal sets used the nonconformity score one minus the softmax probability of the true class, a target miscoverage of 0.1, and the calibration split (about 400 images, which shares source images with the training split, Sec. 3.2.1). The threshold was the mean over 200 bootstrap resamples (80% of the calibration set, with replacement) of the finite-sample-corrected split-conformal quantile at level $\lceil (n+1)(1-\alpha) \rceil / n$; a set contains every class whose softmax probability is at least one minus the threshold. This averaged threshold is a stabilizing heuristic without the finite-sample guarantee of a single split-conformal threshold, which we also evaluate for comparison on the cluster-grouped re-run. Under external shift the exchangeability assumption behind the coverage guarantee does not hold, so we use set size only as an uncertainty signal and make no coverage claim under shift. The combined signal averages the ranks of temperature-scaled maximum softmax confidence and inverted set size. Selective prediction was scored by risk-coverage AUC in distribution, on Herlev (moderate shift), and on Mendeley LBC (severe shift), labeled by the observed degradation of DeiT-Tiny (Table S2).

3.6 Efficiency, Degradation, and AI Use

Parameters and FLOPs (via `torch.utils.flop_counter`, counting a multiply-add as two FLOPs) are architecture properties and were measured without training. CPU latency (single random image, PyTorch eager mode, median of 30 iterations after 5 warm-up iterations) was measured with untrained weights, which is appropriate because dense-inference latency depends on architecture and tensor shapes rather than learned values, on one shared consumer laptop whose CPU model and thread count were not recorded (Table S10). Explanation cost was a forward, backward, and rollout pass against a bare forward pass, both on CPU. Degradation (JPEG compression, Gaussian blur, brightness/contrast shift at five severities) and a synthetic out-of-distribution (OOD) check (25 images of noise, solid colors, and checkerboards) used a freshly trained EfficientViT-M0. Generative AI tools assisted with manuscript drafting and language refinement, under the author's direction, including merging text from the author's two earlier manuscripts and cross-checking reported numbers against source tables.

3.7 Run-to-Run Variation and Statistical Analysis

In-distribution accuracy of the same model differs slightly between original tables (EfficientViT-M0: 0.9654 in Table S1, 0.9737 in Table S2, 0.9621 in Table S3, about 0.975 in the degradation experiment; all five-class). The stages used the same recipe and seed (13) but trained each model afresh in separate sessions, and the training script sets only the PyTorch seed (no deterministic cuDNN; NumPy, Python, and the four shuffled data-loader workers not separately seeded), so identically seeded runs are not expected to be bit-identical. We did not isolate the cause and treat the differences as run-to-run variation of unidentified origin, valid to compare within a table, not across tables. Same-seed repeatability was not measured. Per-image predictions were not retained for the original runs, so bootstrap intervals were not possible for them. The cluster-grouped re-run (Sec. 4.3) retained them, and for it we report 95% bootstrap intervals (2,000 resamples) for balanced accuracy, resampling source clusters for in-distribution and images for external results, plus per-class sensitivity and specificity. For the original runs we report (i) mean and standard deviation over three seeds (Table S4); (ii) Spearman correlations across the 12 models between three-seed mean external balanced accuracy and in-distribution accuracy, parameters, FLOPs, and family (CNN versus transformer; Table 2), reported descriptively because $|\rho| \geq 0.587$ is needed for two-sided $p < 0.05$ without correction for twelve comparisons; and (iii) conservative worst-case 95% half-widths for test-set sampling error of a balanced accuracy, $1.96 \times 0.5\sqrt{0.25/n_{\text{normal}} + 0.25/n_{\text{abnormal}}}$. Exact software versions and hardware are in Table S10.

4 Results

4.1 Benchmark and Cross-Dataset Robustness

The two smallest models, EfficientViT-M0 and EfficientViT-B0 (about 2.1M parameters), were the two most accurate in distribution (Table S1, Fig. S1); EfficientViT-M0 exceeded Swin-Tiny (27.5M parameters), ResNet-50 (23.5M), and ViT-Small (21.7M) by 0.5, 1.1, and 2.8 points. This is a single-seed result on a cell-level split with overlap (Sec. 3.2.1), and across three seeds the two ranked second and third by mean (0.966 each, behind TinyViT-5M at 0.968) with standard deviations of 0.008–0.009, as large as the gaps among the leaders.

In-distribution accuracy did not predict robustness consistently. EfficientViT-B0 (0.9738) and EfficientViT-M0 (0.9737) ranked 9th and 10th of 12 on Herlev and 6th and 10th on Mendeley LBC (Table S2). EfficientViT-M0, the better in-distribution model of a close pair, was worse than DeiT-Tiny (0.9737 versus 0.9504 in distribution) under both shifts (Herlev 0.6716 versus 0.7151; Mendeley LBC 0.4731 versus 0.5230) and on APCEM (0.3730 versus 0.4469; Table S3). On Mendeley LBC all three CNNs scored 0.539–0.667 and eight of nine transformer-family models 0.293–0.523 (the ninth, MobileViTv2, was 0.003 above the lowest CNN, a gap the seed check shows is noise); this split does not hold on Herlev, where transformers hold both the best (MobileViTv2, TinyViT-5M) and the weakest (LeViT-128) positions. MobileViT-S scored far below

chance on Mendeley LBC (balanced accuracy 0.2927, AUROC 0.215), whereas its successor MobileViTv2 was the strongest model on Herlev (0.7501); because Mendeley LBC images are whole fields, this may reflect the scale and content mismatch as much as institutional differences. On APCEM, 9 of 12 models fell below 0.50 on both balanced accuracy and AUROC, and only MobileNetV2 (0.6158, AUROC 0.6528), EfficientNet-Lite0 (0.5393, 0.5701), and LeViT-128 (0.5263, 0.5376) scored above chance. The CNN family spans the best model (MobileNetV2) and the lowest balanced accuracy (ResNet-50, 0.3339), so the CNN advantage seen on Mendeley LBC does not survive a third dataset (Figs. S2 and S3).

4.2 Seed Sensitivity

Variance across seeds is a known threat to benchmark conclusions,[56,57] so all 12 models were repeated at seeds 17 and 29 (Table S4). Close calls can go either way: MobileViTv2's 0.003 edge over MobileNetV2 on Mendeley LBC does not survive (ranges 0.494–0.623 and 0.493–0.604, ranking flipping at seed 17), whereas EfficientViT-B0's 0.001 edge over DeiT-Tiny on APCEM does (it wins at all three seeds, margins 0.027 then 0.067). The Mendeley LBC CNN-versus-transformer split is seed-specific: 1 of 9 transformer models exceeds the weakest CNN at seed 13, 3 of 9 at seed 17, and 4 of 9 at seed 29, by margins up to 0.11. EfficientViT-M0 ranks 10th of 12 on Herlev and Mendeley LBC at seed 13 but mid-pack at seed 29 and is not the in-distribution winner at seed 17; by three-seed mean it ranks second in distribution but 8th, 7th, and 8th of 12 on Herlev, Mendeley LBC, and APCEM, so it is a seed-13 illustration, not a reproducible property. Two findings replicate at all seeds: most transformers outscore CNNs on Herlev, and MobileNetV2 is strongest on APCEM.

FLOPs correlated positively with Herlev accuracy ($+0.68$) but negatively with Mendeley LBC (-0.55) and APCEM (-0.48) accuracy; in-distribution accuracy correlated $+0.55$, -0.31, and -0.27; and family (CNN versus transformer) -0.42, $+0.64$, and $+0.36$ (Table 2). The two correlations above the uncorrected threshold (FLOPs on Herlev, family on Mendeley LBC) would not survive correction and weaken or reverse elsewhere, so we conclude only that these properties did not predict robustness consistently across these datasets. Worst-case 95% sampling half-widths for one balanced accuracy are ± 0.037 (Herlev), ± 0.033 (Mendeley LBC), and ± 0.019 (APCEM) before seed variance, and the seed standard deviations (median 0.025, 0.051, and 0.034) are similar or larger, so many neighboring ranks are indistinguishable.

4.3 Cluster-Grouped Re-Run

To gauge the overlap effect, all 12 models were retrained once (seed 13, same recipe) on a cluster-grouped split in which every source cluster image lies in one split (2,439 training, 630 validation, 384 calibration, and 596 test cells; 163 test clusters), evaluated zero-shot on the external datasets, and per-image predictions were retained (Table 3). It is a single run compared unpaired with the earlier cell-level run (different test cells), so it estimates the effect only roughly.

Table 2: Spearman rank correlation ($n = 12$ models) between three-seed mean external balanced accuracy and static properties. Family is coded CNN = 1, transformer = 0. $|\rho| \geq 0.587$ corresponds to two-sided $p < 0.05$ without correction for multiple comparisons.

Property	Herlev	Mendeley LBC	APCEM
In-distribution accuracy (five-class)	$+0.55$	-0.31	-0.27
Parameters	$+0.43$	-0.46	-0.41
FLOPs	$+0.68$	-0.55	-0.48
Family (CNN versus transformer)	-0.42	$+0.64$	$+0.36$

Table 3: Cluster-grouped re-run (seed 13). The SIPaKMeD column is five-class balanced accuracy with a 95% cluster-bootstrap interval; Δ is the difference from the cell-level seed-13 value in Table S2 (unpaired). External columns are binary balanced accuracy with 95% image-bootstrap intervals. Rows are in the order of Table S2.

Model	SIPaKMeD (5-class)	Δ	Herlev	Mendeley LBC	APCEM
EfficientNet-Lite0	0.953 [0.93, 0.97]	$+0.008$	0.540 [0.51, 0.57]	0.632 [0.60, 0.66]	0.520 [0.50, 0.54]
ResNet-50	0.961 [0.94, 0.98]	$+0.006$	0.721 [0.69, 0.76]	0.541 [0.51, 0.57]	0.516 [0.50, 0.54]
MobileViTv2	0.968 [0.95, 0.98]	$+0.004$	0.722 [0.69, 0.75]	0.487 [0.46, 0.52]	0.410 [0.39, 0.43]
MobileNetV2	0.912 [0.88, 0.94]	-0.038	0.633 [0.60, 0.67]	0.531 [0.50, 0.56]	0.648 [0.63, 0.67]
DeiT-Tiny	0.958 [0.94, 0.98]	$+0.008$	0.730 [0.70, 0.76]	0.460 [0.44, 0.48]	0.434 [0.41, 0.45]
EfficientViT-B0	0.953 [0.93, 0.97]	-0.021	0.625 [0.59, 0.66]	0.559 [0.53, 0.59]	0.426 [0.41, 0.44]
Swin-Tiny	0.951 [0.93, 0.97]	-0.008	0.747 [0.71, 0.78]	0.436 [0.42, 0.46]	0.332 [0.31, 0.35]
ViT-Small	0.935 [0.91, 0.96]	-0.022	0.715 [0.68, 0.75]	0.443 [0.42, 0.47]	0.498 [0.49, 0.51]
LeViT-128	0.807 [0.75, 0.86]	-0.146	0.538 [0.51, 0.57]	0.504 [0.50, 0.51]	0.510 [0.51, 0.51]
EfficientViT-M0	0.947 [0.92, 0.97]	-0.027	0.655 [0.62, 0.69]	0.448 [0.42, 0.47]	0.441 [0.42, 0.46]
TinyViT-5M	0.958 [0.94, 0.98]	-0.010	0.703 [0.67, 0.74]	0.418 [0.39, 0.45]	0.434 [0.43, 0.44]
MobileViT-S	0.963 [0.95, 0.98]	$+0.011$	0.659 [0.63, 0.69]	0.416 [0.39, 0.44]	0.476 [0.47, 0.49]

In distribution, the median change from the cell-level split was -0.9 points of balanced accuracy (range -14.6 to $+1.1$). The largest change, for LeViT-128, is a training failure (early stopping at epoch 8, balanced accuracy 0.807) in a model already unstable across seeds (Table S4: 0.900 ± 0.065), not a leakage effect; without it the median is -0.8 points. The changes are of the size of the seed standard deviations (median 0.5 points) and of the cluster-bootstrap half-widths (about ± 2 points; cluster resampling widened intervals by about 27% over image resampling), so one re-run shows no large inflation from the overlap and does not exclude a small one. External results followed the same pattern as the original runs (Herlev 0.54–0.75; Mendeley LBC and APCEM mostly at or below 0.5), and per-class results show that much of the below-chance performance on the field-level datasets is near one-class prediction: on APCEM ViT-Small had sensitivity 0.07 and specificity 0.93, whereas TinyViT-5M had 0.84 and 0.03. Per-class metrics and confusion matrices are in `runs/kaggle_cluster_split/results.json`.

4.4 Explainability Fidelity

On the original 50 cells (Table S7), relevancy propagation won every comparison, with nucleus concentration 2.71 times chance versus 1.18 for attention rollout, and the ranking replicated on ViT-Small (4.20 versus 1.35; Table S8). Those means came from the cell-level run without per-cell scores, so we repeated the comparison with the cluster-grouped DeiT-Tiny and ViT-Small checkpoints (test balanced accuracy 0.958 and 0.935) on all 596 cluster-grouped test cells, every one with an expert nucleus polygon, using 95% cluster-bootstrap intervals (2,000 resamples of 163 clusters; means in Table S6). Relevancy propagation was better than both baselines on all three metrics on both backbones, all 12 paired intervals excluded zero (Table 4), and it was better on 68–88% of cells; on DeiT-Tiny its nucleus concentration was 3.79 times chance versus 1.35 for attention rollout and 1.19 for Grad-CAM. Representative cells (Fig. 1) show that a high concentration ratio does not always mean attention to the nucleus itself: in some correctly classified cells relevancy propagation concentrates on the nucleus, and in others it highlights the surrounding cytoplasm. We attribute the gap to attention rollout being class-agnostic while relevancy propagation is class-discriminative. Both backbones are plain self-attention transformers, and relevancy propagation is not defined for hybrid architectures such as EfficientViT-M0.

Table 4: Paired per-cell difference, relevancy propagation minus the baseline, mean [95% cluster-bootstrap interval] (percentage of cells on which relevancy propagation is better). Negative deletion differences and positive insertion and concentration differences favor relevancy propagation.

Backbone	Comparison	Deletion AUC	Insertion AUC	Nucleus concentration
DeiT-Tiny	vs. attention rollout	-0.080 [-0.097, -0.063] (88%)	+0.025 [+0.017, +0.033] (74%)	+2.44 [+1.62, +3.29] (78%)
DeiT-Tiny	vs. Grad-CAM	-0.089 [-0.121, -0.059] (76%)	+0.061 [+0.037, +0.086] (68%)	+2.60 [+1.59, +3.62] (76%)
ViT-Small	vs. attention rollout	-0.045 [-0.060, -0.029] (73%)	+0.015 [+0.010, +0.020] (77%)	+2.84 [+1.78, +3.97] (74%)
ViT-Small	vs. Grad-CAM	-0.189 [-0.254, -0.124] (79%)	+0.182 [+0.119, +0.246] (79%)	+3.62 [+2.25, +5.12] (78%)

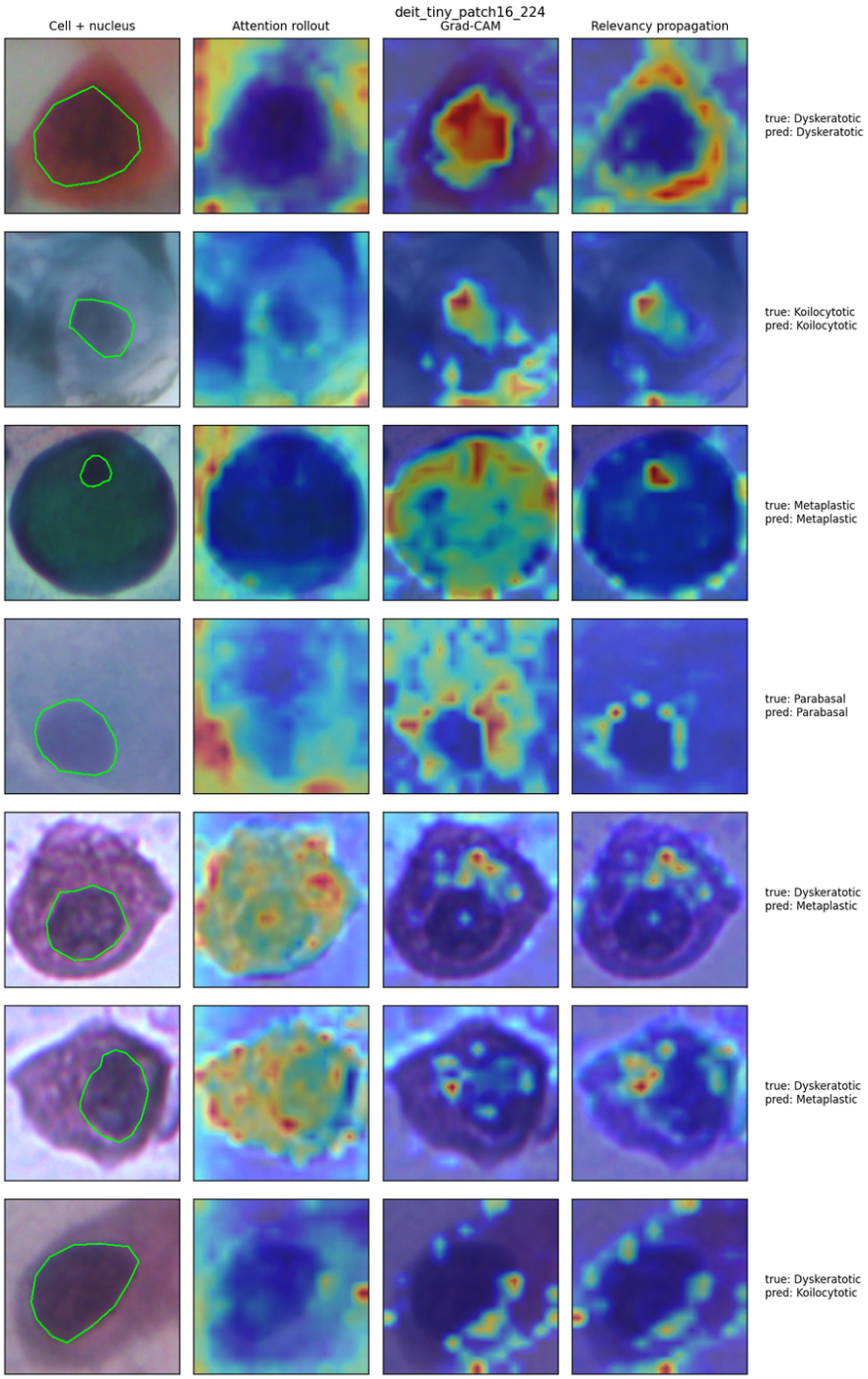


Figure 1: Representative cluster-grouped test cells for DeiT-Tiny: the cell with its expert nucleus outline, and the attention rollout, Grad-CAM, and relevancy propagation heatmaps. The first four rows are the first correctly classified cell of each of the first four classes in test order (the Superficial-Intermediate class is omitted because heatmaps were stored for only the first 400 test cells), and the last three rows are the first misclassified cells; cells were not chosen for appearance. The ViT-Small figure is in the repository (`runs/kaggle_cluster_split_explain/`).

4.5 Selective Prediction

Two sub-claims about safe composition hold by construction in our post-hoc design and were verified: temperature scaling changed zero argmax predictions on the calibration split, and explanation computation is read-only. For selective prediction (Table 5; Fig. S5), confidence alone was the best signal in distribution (0.0050) and on Herlev (0.1190), where combining it with the noisier, discrete set size diluted it (0.0173 and 0.1793), whereas under severe shift on Mendeley LBC the combined signal was best (0.2855 versus 0.3244). Combining therefore helps where the better signal was already struggling and hurts elsewhere, so an evaluation at one shift severity would report the result backwards. The in-distribution values are likely optimistic because of the source-image overlap, and the result rests on one backbone.

Table 5: Selective-prediction risk-coverage AUC (lower is better) on DeiT-Tiny: confidence alone, conformal set size alone, and the combined signal.

Condition	Confidence only	Set size only	Combined
In-distribution	**0.0050**	0.0834	0.0173
Herlev (moderate shift)	**0.1190**	0.3281	0.1793
Mendeley LBC (severe shift)	0.3244	0.4395	**0.2855**

On the cluster-grouped DeiT-Tiny model (384-image calibration split, temperature 1.6), the standard split-conformal threshold gave 89.1% coverage (target 90%) with an average set size of 0.90 on in-distribution test cells, and the bootstrap-averaged threshold gave 89.8% and 0.91, so the stabilizing heuristic changed the sets little. Abstaining on the least confident cases (Table 6) cut in-distribution error from 4.2% to 0.4% at 80% coverage but helped little under cross-dataset conditions (Herlev 22.9% to 19.2%; Mendeley LBC 44.4% to 42.2%), and on APCEM the error rose (64.5% to 66.7%), so confidence was not informative there. These operating points use confidence alone; the combined signal was not recomputed for the re-run.

Table 6: Error rate (%) among accepted cases at fixed coverage for the cluster-grouped DeiT-Tiny model, abstaining on the least confident cases (temperature-scaled maximum probability). In-distribution is five-class; external columns are binary. Lower is better.

Coverage	In-distribution	Herlev	Mendeley LBC	APCEM
100%	4.2	22.9	44.4	64.5
90%	0.6	20.5	43.8	65.8
80%	0.4	19.2	42.2	66.7
70%	0.0	17.9	40.4	68.3

4.6 Efficiency, Explanation Cost, and Degradation

Neither parameter count nor FLOPs reliably predicted CPU latency (Table S5). EfficientNet-Lite0 (3.38M parameters, 0.77 GFLOPs) had more than four times ResNet-50's latency (137.7 versus 32.0 ms) despite ResNet-50 having roughly ten times the FLOPs, and EfficientViT-M0, the leader on parameters and FLOPs, was not the fastest: DeiT-Tiny (15.4 ms) and LeViT-128 (17.5 ms) were faster despite about 14 and 5 times more FLOPs. Latencies carry roughly a 30% run-to-run swing from background load (three isolated EfficientViT-M0 measurements 27.4–28.4 ms; two full sweeps 28.3 and 36.2 ms), but the ordering was stable. On a cloud-VM CPU, a bare DeiT-Tiny forward pass took a median of 49.6 ms and a prediction with a relevancy-propagation explanation 164.1 ms, about 3.3 times the bare cost; absolute figures from the two machines are not comparable, so the ratio is a within-machine comparison. EfficientViT-M0 degraded gradually under all three corruptions (Fig. S6); blur was the most damaging (32.8 points at radius 8, versus 15.5 for JPEG and 18.1 for brightness/contrast), plausibly because Bethesda diagnosis depends on fine nuclear texture. On the 25-image synthetic OOD sample, mean maximum-softmax confidence fell from 0.985 to 0.743, far above the 0.20 five-class chance level, so confidence thresholding alone gives a weak OOD signal.

5 Discussion

The results argue against intuitive heuristics for selecting and verifying a deployment model from one evaluation. "Smaller is worse" was not supported in distribution (with the overlap caveat; a cluster-grouped re-run changed scores only modestly). "The in-distribution leader is the safe choice" was not supported, and even the leader varied with the seed. "Architecture lineage predicts robustness" is contradicted by MobileViT-S and MobileViTv2, and "CNNs are the robust family" looked stable on Herlev and Mendeley LBC but not on APCEM. Across datasets no static property kept the direction of its association with robustness (Table 2), so a pattern that survives two external datasets may not survive a third.

The external results measure something narrower than deployment across institutions. Herlev, the dataset closest in content to the training data, gave the highest external accuracies and a seed-stable pattern in which most transformers outscored CNNs. Mendeley LBC and APCEM ask a single-cell classifier to classify whole fields (APCEM also enhanced), so below-chance results there show that these classifiers do not transfer to field-level images, not how they would behave on single-cell crops from another institution.

The other properties behaved differently from robustness and from each other. Explainability method choice gave a consistent winner on both backbones, supported by paired intervals on 596 cells, although nucleus concentration is not a clinical validity measure. Abstention composition depended on shift severity. Efficiency claims required measurement: FLOPs did not predict latency, and an explanation cost about 3.3 times a bare prediction. No single verification strategy would have characterized all five properties correctly. For deployment, a backbone should not be chosen

from an in-distribution leaderboard; robustness should be measured under the intended shift and across seeds; confidence-based abstention did not reach a reliable operating point under shift, so a screening tool should expect to route many shifted-domain cases to specialists; and focus quality in low-cost capture settings deserves priority because blur was the most damaging corruption.

6 Limitations

Data and external tests. The datasets come from few institutions, and the shifts are cross-dataset differences, not controlled sweeps. They differ in image content (cells versus fields), preprocessing (APCEM enhancement), and label definitions, which cannot be separated here, and binary harmonization discards granularity and rests on category correspondences that the documentation does not establish. Only the observation that static properties were not consistent predictors is likely to generalize; specific rankings should not be assumed to transfer.

Source-image overlap. The in-distribution split is at the cell level with 87.8% overlap and no patient or slide identifiers, so in-distribution accuracy, calibration, abstention, and degradation results may be optimistic. One cluster-grouped re-run changed accuracy by a median of -0.9 points, comparable to seed variation, but the comparison is unpaired and uses one split; a valid estimate needs a paired design, such as repeated cell-level and cluster-grouped cross-validation on the same cells with variability across which clusters fall in the test set.[22,23]

Seeds, runs, and statistics. Headline results are single runs at seed 13, and the seed check showed that several claims are seed-specific. Original-run tables come from separate runs of unidentified variation, same-seed repeatability was not measured, and per-image predictions and checkpoints were not retained for the original runs, so intervals and per-class metrics exist only for the cluster-grouped re-run, where the combined-signal abstention was not repeated. Exact software versions were not recorded for the original runs (Table S10).

Backbones and measurement. The five properties were not measured on one backbone (benchmark and efficiency: 12 models; explainability and abstention: DeiT-Tiny, with ViT-Small for explainability; degradation and OOD: EfficientViT-M0), so we did not test whether a single backbone would change the central observation, and the results may not transfer to hybrid architectures or other imaging tasks. The explainability analysis covers two related plain transformers, one deletion/insertion configuration (20 steps, blurred baseline, which can affect rankings), and one run. The OOD check is a 25-image synthetic sample. CPU latency was measured on a shared laptop with about 30% run-to-run swing, so the 3.3-times explanation multiplier is approximate.

Clinical validity. The nucleus concentration ratio measures overlap with an annotated nucleus, not clinical validity or reader agreement, and post hoc explanations offer limited assurance about a model's reasoning unless validated.[58] No expert review, reader study, or prospective evaluation was performed.

7 Conclusion

Five properties of single-cell cervical cytology classification, tested across experiments on a common dataset family and model registry, did not resolve alike. Parameter-efficient ViTs did not trade accuracy for efficiency in distribution, yet the in-distribution leaderboard misled about robustness: no static property predicted robustness consistently across the three external datasets, and apparent family-level patterns did not survive a third dataset or a change of seed. In-distribution scores came from a cell-level split with source-image overlap, although a cluster-grouped re-run changed them only modestly. Relevancy propagation was more faithful than attention rollout or Grad-CAM on both evaluated transformer backbones, combining confidence with conformal set size helped only under the severest shift, and efficiency claims held only under direct measurement. Each property needed its own test, under the conditions of intended deployment.

Disclosures

The author declares that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code, Data, and Materials Availability

All datasets analyzed are publicly available: SIPaKMeD,[48] Herlev,[49] Mendeley LBC,[50] and APCEM (`https://doi.org/10.17632/snzb5g6ddy.2`).[51] The training and evaluation code and the per-model result files are available from the corresponding author upon reasonable request.

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